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# MiSTer Monitor

## Setup Guide

v2.7.0

### RETROACHIEVEMENTS

PROGRESS 23/40

57%



|     |                   |     |
|-----|-------------------|-----|
| [H] | Perfect Run       | 25p |
| [*] | First Blood       | 10p |
| [ ] | Speed Demon       | 50p |
| [ ] | No Continues Used | 25p |

#### ACHIEVEMENT!

+25 pts Perfect Run



## What is MiSTer Monitor

MiSTer Monitor turns an ESP32-based touchscreen into a companion display for your MiSTer: the moment you load a game, the screen shows its box art, a “Now Playing” panel with related info (year, dev, publisher, genre, players, rating and synopsis) of the active game, your **RetroAchievements** progress, and live system stats. Everything is fetched on demand.

Two pieces work together: a tiny **server** on the MiSTer that watches what you play, and the **firmware** on the display that renders it. You will install both.

### You will need

- A **MiSTer** connected to your home network.
- A supported display: **M5Stack Tab5** or a **Cheap Yellow Display** (2.8" ILI9341/ST7789, or 3.5" capacitive/resistive).
- A **microSD card** for the display.
- A **USB data cable** and a desktop computer with **Chrome or Edge** (only for the one-time flash).
- A free **ScreenScraper** account — created in step 3.
- Optional: a free **RetroAchievements** account — step 5.

## 1

### Flash the display

- Connect the display to your computer with the USB cable.
- Open the flasher page in **Chrome or Edge** (Firefox and Safari don't support Web Serial):

```
https://chipster6502.github.io/MiSTer\_monitor/flasher/
```

- Click **Connect** next to your board, pick the serial port in the browser dialog, and wait for the confirmation.
- Not sure which CYD variant you have? The page itself explains how to tell them apart.

## 2

### Prepare the display's microSD card

The repository ships a ready-made card layout under **SD\_card\_content/**. Pick the subfolder matching your board, copy its contents to the **root** of the microSD card, then open **config.ini** in any text editor and fill in your network — **without quotes** around the values:

```
config.ini — on the display's microSD card
[wifi]
ssid=YOUR_WIFI_NAME
password=YOUR_WIFI_PASSWORD
```

That is all the display needs to find the MiSTer: it discovers it automatically on the network at boot.

**NOTE** Leave `[mister] ip=` blank. Only set it if your router blocks UDP broadcast, which is rare in home networks.

### 3 Create a free ScreenScraper account

- Register at **screenscraper.fr** — free, instant, no approval needed.
- Add your username and password to the same **config.ini**:

```
config.ini – same file, next section  
[screenscraper]  
ss_user=YOUR_SS_USERNAME  
ss_pass=YOUR_SS_PASSWORD
```

The app brings its own developer credentials, so a plain member account is enough. Artwork is downloaded the first time each game appears and cached on the card, so it costs quota once and loads instantly forever after.

### 4 Install the server on the MiSTer

Through the MiSTer Downloader — updates itself from then on

- Download the drop-in database file and place the **.ini** inside it in the root of the MiSTer SD card (**/media/fat/**):

```
github.com/chipster6502/MiSTer\_monitor\_DB  
→ downloader\_chipster6502\_MiSTer\_monitor\_DB.zip
```

Direct link:

[raw.githubusercontent.com/chipster6502/MiSTer\\_monitor\\_DB/db/downloader\\_chipster6502\\_MiSTer\\_monitor\\_DB.zip](https://raw.githubusercontent.com/chipster6502/MiSTer_monitor_DB/db/downloader_chipster6502_MiSTer_monitor_DB.zip)

- Run **Update All** (or **Downloader**) from the Scripts menu — it installs the server files.
- Back in the Scripts menu, run **MiSTer\_Monitor\_setup** once. It enables auto-start, fixes **MiSTer.ini** for game detection, and starts the server right away.
- Insert the card in your display and power it.
- Load any game. Within a few seconds the display shows its artwork.

**TIP** From now on, every **Update All** keeps the server current automatically. To pause the monitor, run **MiSTer\_Monitor\_uninstall**; run the setup script again to re-enable it.

### 5 RetroAchievements (optional)

Your progress and live unlock popups on the display

This part lives **on the MiSTer**, not on the display's card. The first time the server runs it drops a ready-to-edit file in place — you only fill in two lines:

```

on the MiSTer — created automatically on first start
/media/fat/Scripts/.config/mister_monitor/ra_credentials.ini

[retroachievements]

username=YourRAUsername

api_key=YourWebAPIKey

```

Your **Web API key** is at [retroachievements.org](https://retroachievements.org) → *Settings* → *Applications*. Load a game with a published set and the RetroAchievements page comes alive: progress bar, points, and the full trophy list — tap any row for its description.

| Your setup                              | What you get                                                                                                                     |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Credentials only<br>stock MiSTer        | The full page in <b>view-only</b> mode: the game's set, your cloud progress and every trophy — but play is not recorded.         |
| odelot/Main_MiSTer fork<br>RA_cores     | Achievements are <b>earned while you play</b> ; a READY banner confirms it on each matched game, and unlocks pop within seconds. |
| Fork + debug=1<br>retroachievements.cfg | Popups become <b>instant</b> (with title and description), and CD systems — PlayStation, Saturn, Mega CD — resolve too.          |

**IMPORTANT** Each tier is optional and falls back gracefully to the one below. Without the credentials file, everything else in this guide works exactly the same.

## Earning achievements on the MiSTer

Recording achievements needs odelot's RetroAchievements build of the MiSTer main binary, plus his modified core for each system. Two maintained routes install it for you, both keeping your stock cores intact:

- **Update All** — the RetroAchievements Cores database, maintained by odelot himself, is now built in. Run **Update All**, enable it in the settings screen, and it walks you through the `retroachievements.cfg` setup and the `MiSTer.ini` entries it needs.
- **MiSTer Companion** — install the **RetroAchievements Cores** entry from its **Install Center**.

Either route leaves a `retroachievements.cfg` on the MiSTer holding your RA **username and password**. That file belongs to odelot's build and is separate from the `ra_credentials.ini` above, which is this project's and takes an API key instead — you will end up with both, and that is expected.

```

two different files, both needed for the full experience

/media/fat/retroachievements.cfg      odelot's build  (user + password)
/media/fat/Scripts/.config/mister_monitor/ra_credentials.ini
                                     this project    (user + API key)

```

**IMPORTANT** Once the RA cores are installed an **RA\_Cores** entry appears in the MiSTer OSD. Achievements are only active for cores launched from there — opening the same system from its usual menu entry runs the stock core and records nothing.

**TIP** Add `debug=1` to `retroachievements.cfg` for sub-second unlock popups. Achievements only work on the systems odelot has published cores for — his repository lists the current set.



## You're done — a quick check

### Working as intended looks like this

- Boot: the display connects to WiFi and finds the MiSTer by itself.
- Load a game: box art appears within seconds; the Game Info panel fills in year, developer and synopsis.
- RetroAchievements page: progress and trophies for the running game (view-only unless you installed the fork and open a RA enabled core).
- System pages: live CPU, memory, storage and network stats.

```
links
Project      github.com/chipster6502/MiSTer_monitor
Docs         installation.md    configuration.md
Flasher      chipster6502.github.io/MiSTer_monitor/flasher/
RetroAchiev. retroachievements.org
RA fork      github.com/odelot/Main_MiSTer
Update All   github.com/theypsilon/Update_All_MiSTer
Companion    mistercompanion.org
```

MiSTer Monitor is free software — firmware under GPL-3.0, server under AGPL-3.0. Found a rough edge in this guide? Open an issue; it helps the next person.